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CONTROL PANEL LOAD LIST

200 KLD Sewage Treatment Plant • KVG Ayurveda Medical College, Sullia

Client / Project	KVG Ayurveda Medical College	Panel	STP Main Control Panel (MCC cum PLC)
System	200 KLD Sewage Treatment Plant (STP)	Supply	415 V ±10%, 3Ph, 4W, 50 Hz
Location	Sullia	Control Voltage	230 V AC / 24 V DC

CONNECTED LOAD (HP)	CONNECTED LOAD (kW)	OPERATING LOAD (HP)	OPERATING LOAD (kW)
51.25	38.24	29.25	21.83

A. POWER OUTPUTS – MOTOR & ELECTRICAL LOADS

Sl. No.	Equipment Description	Rating (HP)	Rating (kW)	Supply	Qty (Nos.)	Duty	Connected Load (HP)	Connected Load (kW)	Operating Load (kW)
1	Feed Pump (Working)	2.00	1.49	415V, 3Ph	1	Working	2.00	1.49	1.49
2	Feed Pump (Standby)	2.00	1.49	415V, 3Ph	1	Standby	2.00	1.49	–
3	Permeate Pump (Working)	2.00	1.49	415V, 3Ph	1	Working	2.00	1.49	1.49
4	Permeate Pump (Standby)	2.00	1.49	415V, 3Ph	1	Standby	2.00	1.49	–
5	Backwash Pump	1.00	0.75	415V, 3Ph	1	Working	1.00	0.75	0.75
6	Solenoid Valve – SV1	0.25	0.19	230V, 1Ph	1	Working	0.25	0.19	0.19
7	Solenoid Valve – SV2	0.25	0.19	230V, 1Ph	1	Working	0.25	0.19	0.19
8	Air Blower (Working)	15.00	11.19	415V, 3Ph	1	Working	15.00	11.19	11.19
9	Air Blower (Standby)	15.00	11.19	415V, 3Ph	1	Standby	15.00	11.19	–
10	RAS Pump (Submersible)	2.00	1.49	415V, 3Ph	1	Working	2.00	1.49	1.49
11	Treated Water Pump (Working)	3.00	2.24	415V, 3Ph	1	Working	3.00	2.24	2.24
12	Treated Water Pump (Standby)	3.00	2.24	415V, 3Ph	1	Standby	3.00	2.24	–
13	Extra Pump – 1	1.00	0.75	415V, 3Ph	1	Working	1.00	0.75	0.75
14	Extra Pump – 2	2.00	1.49	415V, 3Ph	1	Working	2.00	1.49	1.49
15	Mechanical Screen	0.50	0.37	415V, 3Ph	1	Working	0.50	0.37	0.37
16	Dosing Pump	0.25	0.19	230V, 1Ph	1	Working	0.25	0.19	0.19
TOTAL					16		51.25	38.24	21.83

B. FIELD INSTRUMENTS – ANALOG / COMMUNICATION INPUTS

Sl. No.	Instrument Description	Signal Type	Supply	Qty (Nos.)
1	Vacuum Transmitter	4–20 mA	24V DC	1
2	Pressure Transmitter	4–20 mA	24V DC	1
3	pH Meter	4–20 mA	230V, 1Ph	1
4	Electromagnetic Flow Meter (EMF)	RS-485 (Modbus)	230V, 1Ph	1

C. DIGITAL INPUTS

Sl. No.	Input Description	Signal Type	Supply	Qty (Nos.)
1	Float Level Switch – LS1	Potential-free (NO/NC)	24V DC	1
2	Float Level Switch – LS2	Potential-free (NO/NC)	24V DC	1
3	Float Level Switch – LS3	Potential-free (NO/NC)	24V DC	1
4	Float Level Switch – LS4	Potential-free (NO/NC)	24V DC	1
5	Float Level Switch – LS5	Potential-free (NO/NC)	24V DC	1
6	Float Level Switch – LS6	Potential-free (NO/NC)	24V DC	1

D. PANEL ACCESSORIES / ADDITIONAL REQUIREMENTS

Sl. No.	Description	Type	Supply	Qty (Nos.)
1	Panel Illumination Lamp with Door Limit Switch	LED	230V, 1Ph	1
2	Voltmeter with Selector Switch	Digital	415V, 3Ph	1
3	Energy Meter	Digital, RS-485	415V, 3Ph	1
4	3-Pin Switch Socket	6/16 A	230V, 1Ph	2

NOTES

- Conversion factor: 1 HP = 0.7457 kW.
- Operating load includes only equipment with Duty = 'Working'; Standby loads are excluded.
- Motor ratings to be verified against final equipment datasheets before panel manufacturing.